1. (a) (i) The geometric way of seeing this is to derive the convolution $v[n]$ by flipping $y[n]$ and then sliding it along $x[n]$. We can divide the convolution into three portions:

I. part of $y[n]$ extends before $x[0]$,

II. $y[n]$ fits entirely within $x[n]$,

III. part of $y[n]$, extends past $x[N-1]$.

Only when $y[n]$ fits entirely within $x[n]$ will $v[n] = w[n]$. This corresponds to $M-1 \le n \le N-1$.

To derive this algebraically, we note that for $v[n] = w[n]$, we need $y[n-r] = y[(n-r)_{\text{mod } N}]$ for the summing range $r = 0, \dots, N-1$. We know that $y[i] = y[i_{\text{mod } N}]$ for $0 \le i \le N-1$ since the "mod N " then has no effect. Less obviously, it will also be true when $M-N \le i \le -1$ since for these values $y[i] = y[i+N] = 0$. So overall $y[i] = y[i_{\text{mod } N}]$ whenever $M-N \le i \le N-1$.

So, when $r = 0$, $y[n-r] = y[(n-r)_{\text{mod } N}]$ requires $n-r = n \le N-1$. When $r = N-1$, $y[n-r] = y[(n-r)_{\text{mod } N}]$ requires $n-r = n-N+1 \ge M-N \Longrightarrow n \ge M-1$. Combining these gives $M-1 \le n \le N-1$. [3]

Very few got this right.

(ii) $v[1] = 1\times 2 + 2\times 5 = 12$ and $w[1] = 1\times 2 + 2\times 5 + 3\times 7 = 33$.
For completeness, $v[n] = [5,12,22,16,24,26,21]$ and $w[n] = [31,33,22,16,24]$. [2]

A surprisingly large number of people couldn't do these convolutions correctly.

(b) (i) An LTI system is BIBO stable if a bounded input sequence, $x[n]$ always gives a bounded output sequence $y[n]$. That is, $|x[n]| < B\ \forall n \Rightarrow |y[n]| < f(B)\ \forall n$. [2]

Many people quoted equivalent conditions from the notes but did not actually define what BIBO means.

(ii) Define $x[n] = +1$ if $h[-n] \ge 0$ and $x[n] = -1$ otherwise. Then $y[0] = \sum_{r=-\infty}^{\infty} h[r]x[0-r] = \sum_{r=-\infty}^{\infty} |h[n]|$ which is finite from the BIBO assumption since $x[n] \le 1$ and thus is bounded. [3]

A fairly good number of students answer question correctly. You can only prove the BIBO property by defining an input sequence $x[n]$ that is bounded and then showing that each output sample $x[n]$ is bounded.

(c) (i) $W = 0$ when the numerator is zero but the denominator is non-zero which is when $0.5(M+1)\omega = k\pi$ for $k \ne 0$. This happens first when $k = 1$ giving $\omega = \frac{2\pi}{M+1}$. [1]

(ii) Using the small angle formula $\sin\theta = \theta$ gives $W = \frac{0.5(M+1)\omega}{(M+1)\times 0.5\omega}$ dB for small ω (including ω=0). Equivalently, you can use L'Hôpital's rnule. [2]

(iii) At $\omega = \frac{3\pi}{M+1}$, $|W| = \frac{1}{(M+1)\sin\frac{1.5\pi}{M+1}} \approx \frac{1}{1.5\pi} = 0.212 = -13.5\text{dB}$. [2]

A fairly good number of students solved this question correctly. A variety of flaws in various stages of the calculations were observed.

(d) (i) $H(e^{j\omega}) = \sum_{n=0}^{M} h[n]e^{-j\omega n} = \sum_{n=0}^{\frac{M-1}{2}} h[n]e^{-j\omega n} + \sum_{n=\frac{M+1}{2}}^{M} h[n]e^{-j\omega n}$.

Substituting $r = M - n$ in the second summation and reversing the order of summation gives:

$$H(e^{j\omega}) = \sum_{n=0}^{\frac{M-1}{2}} h[n]e^{-j\omega n} + \sum_{r=0}^{\frac{M-1}{2}} h[M-r]e^{j\omega(M-r)}$$

Since $h[M-r] = h[r]$ we can combine the sums to give

$$H(e^{j\omega}) = \sum_{n=0}^{\frac{M-1}{2}} h[n](e^{-j\omega n} + e^{j\omega(M-r)}) = e^{-\frac{j\omega M}{2}} \sum_{n=0}^{\frac{M-1}{2}} h[n]\left(e^{-j\omega\left(n-\frac{M}{2}\right)} + e^{j\omega\left(n-\frac{M}{2}\right)}\right)$$

$$H(e^{j\omega}) = 2e^{-\frac{j\omega M}{2}} \sum_{n=0}^{\frac{M-1}{2}} h[n]\cos\left(n-\frac{M}{2}\right)\omega$$

Hence, $A = 2$ and $\theta(\omega) = -\frac{M}{2}\omega$. [4]

(ii) $\angle H(e^{j\omega}) = \theta(\omega) = -\frac{M}{2}\omega$ so $\tau_H(e^{j\omega}) = \frac{M}{2}$ [1]

A fairly good number of students solved this question correctly.

(e) From the block diagram we can write $Y = (z^{-1} + a)V$ and $V = X - az^{-1}V$. From the second of these, we get $X = (z^{-1} + a)V$ and combining the two equations (to eliminate V) then gives $\frac{Y}{X} = \frac{z^{-1}+a}{1+az^{-1}}$ which is an allpass filter. [5]

A large number of students solved this question correctly.

(f) (i) For the causal signal we have:

$$X[z] = \sum_{n=-\infty}^{\infty} a^n u[n]z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} \left(\frac{a}{z}\right)^n = 1 + \left(\frac{a}{z}\right) + \left(\frac{a}{z}\right)^2 + \left(\frac{a}{z}\right)^3 + \cdots$$

We apply the geometric progression formula:

$$1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x},\ |x| < 1$$

Therefore, $X[z] = \frac{1}{1-\frac{a}{z}} = \frac{z}{z-a},\ \left|\frac{a}{z}\right| < 1,\ |z| > |a|$

For the anticausal signal $-\gamma^n u[-n-1]$, we have

$$X[z] = \sum_{n=-\infty}^{\infty} -\gamma^n u[-n-1]z^{-n} = \sum_{n=-\infty}^{-1} -\gamma^n z^{-n} = -\sum_{n=1}^{\infty} \gamma^{-n} z^n = -\sum_{n=1}^{\infty} \left(\frac{z}{\gamma}\right)^n$$

$$= -\frac{z}{\gamma}\sum_{n=0}^{\infty} \left(\frac{z}{\gamma}\right)^n = -\left(\frac{z}{\gamma}\right)\left[1 + \left(\frac{z}{\gamma}\right) + \left(\frac{z}{\gamma}\right)^2 + \left(\frac{z}{\gamma}\right)^3 + \cdots\right]$$

Therefore, $X[z] = -\left(\frac{z}{\gamma}\right)\frac{1}{1-\frac{z}{\gamma}} = \frac{z}{z-\gamma},\ \left|\frac{z}{\gamma}\right| < 1,\ |z| < |\gamma|$ [2]

(ii) We wish to write $H[z] = \frac{b}{2-z^{-1}} + \frac{c}{1+4z^{-1}} = \frac{(b+2c)+(4b-c)z^{-1}}{(2-z^{-1})(1+4z^{-1})}$.
By matching coefficients, we obtain $b + 2c$=2 and $4b - c = 17$ from which $b = 4$ and $c = -1$.

$$H[z] = \frac{2}{2 - z^{-1}} - \frac{1}{1 + 4z^{-1}}$$

The corresponding poles are $z = 0.5$ and $z = -4$ so the sequence we need is $2 \times 0.5^n u[n] - (-(-4)^n u[-n-1]) = 2^{1-n}u[n] + (-4)^n u[-n-1]$. [3]

A good number of students used the right approach to solve this question. However, the number and variety of careless mistakes in the calculations was notorious.

(g) (i) Taking z-transforms: $Y(z) = az^{-1}Y(z) + (1-a)X(z)$ from which $H[z] = \frac{Y[z]}{X[z]} = \frac{1-a}{1-az^{-1}}$.
$h[-1] = 0, h[0] = 1 - a, h[1] = (1-a)a, h[2] = (1-a)a^2$. [2]

A fairly good number of students used the right approach to solve this question. However, a large number of students got $h[-1]$ wrong.

(ii) The system function is $H[z] = \frac{1-a}{1-az^{-1}} = \frac{(1-a)z}{z-a}$. This has a pole at $z = a$ and a zero at $z = 0$. [2]

A fairly good number of students answered this question correctly. It is surprising to see how many students still don't know what is the definition of poles and zeros.

(iii) The system function is $\left|H\left(e^{j\omega}\right)\right|^2 = H\left(e^{j\omega}\right) \times H^*\left(e^{j\omega}\right) = 0.5$. Thus

$$0.5 = \frac{1-a}{1-ae^{-j\omega}} \times \frac{1-a}{1-ae^{-j\omega}} = \frac{(1-a)^2}{1-2a\cos\omega + a^2}$$

From this, $\cos\omega = \frac{1+a^2-2(1-a)^2}{2a} = \frac{4a-1-a^2}{2a} = 1 - \frac{(1-a)^2}{2a}$ and so
$\omega_{3dB} = \cos^{-1}\left(\frac{4a-1-a^2}{2a}\right) = \cos^{-1}\left(1 - \frac{(1-a)^2}{2a}\right)$ [3]

A good number of students used the right approach to solve this question. However, the number and variety of careless mistakes and miscalculations were fairly large.

(iii) For $n \geq 0$, we have from part (i) that $(1-a)a^n$. This can also be proved by induction from the given recurrence relation assuming that $x[n]$ is an impulse at $n = 0$. If we now substitute $t = \frac{n}{f_s}$ (or, equivalently $t = nT$ where the sample period is $T = \frac{1}{f_s}$) into $g(t) = Ae^{-\frac{t}{\tau}}$ we obtain $h[n] = g\left(\frac{n}{f_s}\right) = Ae^{-\frac{n}{\tau f_s}}$ from which $A = 1 - a$ and $a = e^{-\frac{1}{\tau f_s}}$. Rearranging the later equation gives $\tau = \frac{-1}{f_s \ln a}$. [3]

A good number of students used the right approach to solve this question and produced correct results.

2. (a) (i) We can write

$$1+\frac{(|\hat{z}|^2-1)(1-\lambda^2)}{|1-\lambda\hat{z}|^2}=\frac{(1-\lambda(\hat{z}+\hat{z}^*)+\lambda^2|\hat{z}|^2)+|\hat{z}|^2-\lambda^2|\hat{z}|^2-1+\lambda^2}{|1-\lambda\hat{z}|^2}$$

$$=\frac{|\hat{z}|^2-\lambda(\hat{z}+\hat{z}^*)+\lambda^2}{|1-\lambda\hat{z}|^2}=\frac{|\hat{z}|^2-\lambda(\hat{z}+\hat{z}^*)+\lambda^2}{|1-\lambda\hat{z}|^2}=\frac{(\hat{z}-\lambda)(\hat{z}^*-\lambda)}{|1-\lambda\hat{z}|^2}$$

$$=\frac{|\hat{z}-\lambda|^2}{|1-\lambda\hat{z}|^2}=|z|^2$$

Since $|\lambda|<1$, the numerator term $(1-\lambda^2)$ must be strictly positive. In addition, the denominator term satisfies $|1-\lambda\hat{z}|^2\geq 0$. Hence, assuming for the moment that $1-\lambda\hat{z}\neq 0$, the sign of the fraction is equal to the sign of $(|\hat{z}|^2-1)$ and is positive or negative according to whether $|\hat{z}|>1$ or $|\hat{z}|<1$. Clearly $|\hat{z}|=1$ makes the fraction zero and hence $|z|=1$. Putting all this together, we have shown that $|\hat{z}|<1\Rightarrow|z|<1$ and $|\hat{z}|\geq 1\Rightarrow|z|\geq 1$ which is equivalent to $|z|<1\Rightarrow|\hat{z}|<1$. The special case, $1-\lambda\hat{z}=0$, arises when $\hat{z}=\lambda^{-1}>1$. In this case, the numerator of the fraction is strictly positive and $|z|=+\infty\not<1$ so the proposition is satisfied. [4]

It is surprisingly that many students have weak knowledge regarding properties of complex numbers. Some people assumed that $z^*=z^{-1}$; this is only true if $|z|=1$ which cannot be assumed for this question. The phrase "if and only if" means that you must prove the implication in both directions; quite a few only proved it one way. Very few people considered the case when the denominator is zero.

(ii) The property implies that the unit circle maps into itself; this means that if the bilinear transformation is used to transform a filter, the frequency response of a transformed filter is the same as that of the original filter but with a distorted frequency axis. A filter is stable iff all its poles lie strictly inside the unit circle. If this transformation is applied to a stable filter, the property proved in part i) ensures that the transformed filter is also stable. In the same way, it also ensures that a minimum phase filter will transform into another minimum phase filter and that a causal filter will transform into a causal filter. [2]

Surprisingly few people correctly stated any of the above properties. Several just stated that the poles of a stable filter had to be inside the unit circle; this is true but is nothing to do with the transformation. Some students made stronger (but untrue) statements such as "the region of convergence is unchanged".

(b) (i) For $\omega=0$, the filter gain is $G(e^{j\omega})=1+(e^{j\omega})^{-1}=2$. At $\omega_G=\frac{\pi}{2}$, the filter gain is $G(e^{j\omega_G})=1+(e^{j\omega_G})^{-1}=1+(j)^{-1}=1-j$. Hence $|G(e^{j\omega_G})|=\sqrt{2}=\frac{G(1)}{\sqrt{2}}$. [2]

A fairly good number of students answered this question correctly.

(ii) For $z=e^{j\omega}$, we can write $z^{1/2}G(z)=z^{1/2}+z^{-1/2}$ and $e^{j\omega/2}G(e^{j\omega})=e^{j\omega/2}+e^{-j\omega/2}=2\cos(\omega/2)$.
Taking the magnitude of each side gives $|G(e^{j\omega})|=2\cos(\omega/2)$ for $|\omega|\leq\pi$.
The dashed line shows $|G(e^{j\omega_G})|=\sqrt{2}$. [4]

A fairly good number of students answered this question correctly.

(iii) We want a lowpass-to-lowpass transformation with $\omega_0 = \frac{\pi}{2}$ and $\hat{\omega}_1 = 0.2$. So

$$\lambda = \frac{\sin\left(\frac{\omega_0 - \hat{\omega}_1}{2}\right)}{\sin\left(\frac{\omega_0 + \hat{\omega}_1}{2}\right)} = \frac{\sin 0.6854}{\sin 0.8854} = \frac{0.633}{0.774} = 0.8176$$

Substituting for

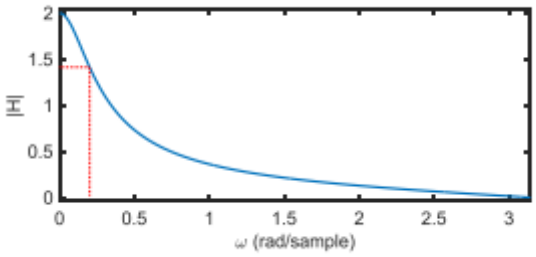
$$H(\hat{z}) = 1 + \frac{\hat{z}^{-1} - \lambda}{1 - \lambda\hat{z}^{-1}} = \frac{1 - \lambda\hat{z}^{-1} + \hat{z}^{-1} - \lambda}{1 - \lambda\hat{z}^{-1}} = (1 - \lambda)\frac{1 + \hat{z}^{-1}}{1 - \lambda\hat{z}^{-1}}$$

$$= 0.1824\frac{1 + \hat{z}^{-1}}{1 - 0.8176\hat{z}^{-1}}$$

[5]

A quite large number of students failed in giving the perfect answer. A variety of flaws and miscalculations were observed here.

(iv) The frequency axis has been distorted non-linearly but the gains at $\omega = \{0, \pi\}$ are preserved.



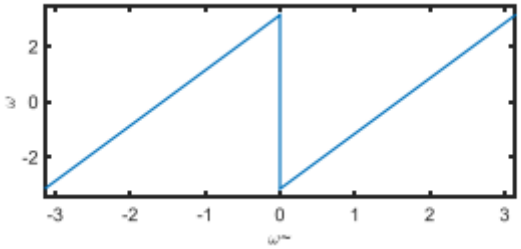
[2]

A fairly good number of students provided a correctly shaped curve but not very accurate.

(c) (i) We see that $|\tilde{z}| < 1 \Leftrightarrow |\tilde{z}|^2 < 1 \Leftrightarrow |\tilde{z}^2| < 1 \Leftrightarrow |-\tilde{z}^2| < 1 \Leftrightarrow |z| < 1$. Hence the transformation preserves stability. [2]

A fairly good number of students answered this question correctly.

(ii) If $z = -\tilde{z}^2$ then $e^{j\omega} = -e^{j2\tilde{\omega}} = e^{j\pi} \times e^{j2\tilde{\omega}} = e^{j(2\tilde{\omega}+\pi)}$ from which $\omega = (2\tilde{\omega} + \pi)_{\text{mod}2\pi}$ (or equivalently $\omega = (2\tilde{\omega} - \pi)_{\text{mod}2\pi}$). [2]

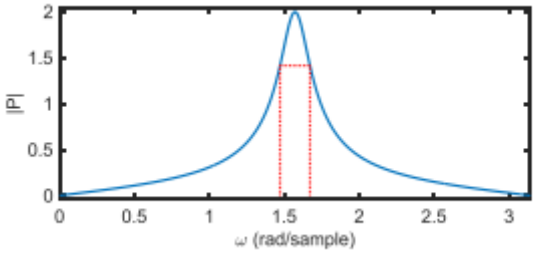


A fairly good number of students provided a correctly shaped curve but not very accurate.

(iii) We have $H(z) = (1-\lambda)\frac{1+z^{-1}}{1-\lambda z^{-1}}$. Making the substitution $z^{-1} = -\tilde{z}^{-2}$ gives $P(\tilde{z}) = (1-\lambda)\frac{1-\tilde{z}^{-2}}{1+\lambda\tilde{z}^{-2}} = 0.1824\frac{1-\tilde{z}^{-2}}{1+0.817\tilde{z}^{-2}}$. [3]

A quite large number of students failed in giving the perfect answer. A variety of flaws and miscalculations were observed here.

(iv)



$\left|H\left(e^{j\omega}\right)\right| = \sqrt{2}$ for $\omega = \pm 0.2$.Hence, $\left|P\left(e^{j\tilde{\omega}}\right)\right| = \sqrt{2}$ when $(2\tilde{\omega}+\pi)_{\mathrm{mod}2\pi} = \omega = \pm 0.2$. Solving this equation gives

$2\tilde{\omega} + \pi = \pm 0.2 + 2n\pi$

$\tilde{\omega} = \pm 0.1 + \left(n - \frac{1}{2}\right)\pi = \cdots, -\frac{3\pi}{2} \pm 0.1, -\frac{\pi}{2} \pm 0.1, \frac{\pi}{2} \pm 0.1, \frac{3\pi}{2} \pm 0.1, \ldots$ [2]

A fairly good number of students provided a correctly shaped curve but not very accurate.

(v)

The two values of ω in the range $0 \le \omega \le \pi$ for which $\left|P\left(e^{j\tilde{\omega}}\right)\right| = \sqrt{2}$ are therefore, $\omega = \frac{\pi}{2} \pm 0.1 = \{1.4708, 1.6708\}$. This is illustrated by the dotted line on the graph. The bandwidth of the filter is 0.2 and is equal to ω_H, the cutoff frequency of $H(z)$. [2]

Only a few students provided an accurate answer here.